Chaoan Li

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Education

Texas A&M University — College Station, TX
Ph.D. Student, Industrial Engineering, Data Science Track · GPA: 4.0/4.0 — *Aug. 2024 – May 2028 (expected)*

Beijing Normal University — Beijing, China
M.S., Mathematics · GPA: 3.7/4.0 — *Sep. 2021 – Jul. 2024*

University of Science and Technology Beijing — Beijing, China
B.S., Applied Mathematics · Minor: Financial Engineering · GPA: 3.94/4.0 — *Sep. 2017 – Jul. 2021*

Technical Skills

Machine Learning: Kernel methods, decision trees, SVM; Transformer, Mamba, CNN, RNN, LSTM; LLM evaluation, text generation, watermark privacy
Statistics: Statistical inference, hypothesis testing, uncertainty quantification, Monte Carlo, experimental design
Mathematical Methods: Functional analysis, variational methods, stochastic processes, high-dimensional statistics
Programming: Python, C#, C++, R, MATLAB, SQL
Libraries & Tools: Git, PyTorch, scikit-learn, NumPy, pandas, SciPy, Jupyter, Linux, Bash; Claude Code, Codex

Research Experience

Texas A&M University — College Station, TX
Research Assistant (Advisor: Prof. Rui Tuo) — *Aug. 2024 – Present*

AI-Generated Text Privacy: Detecting Watermark Key Reuse
- Designed keyless tests of watermark-key reuse within text groups. Python/PyTorch experiments on Gemma-2-2B and Qwen2.5-1.5B **outperformed four text-similarity baselines in short-context settings**; analyzed topic effects and key rotation.

Simulation Reliability: Detecting and Localizing Model Errors
- Developed KRR-based frequency-domain tests and a NumPy/SciPy package for global, regional and post-calibration checks. Found a shear-layer discrepancy **missed by two baseline tests**; assessed false alarms and power with 1,000 simulations per scenario.

Nonlinear Model Calibration and Uncertainty Quantification
- Unified three fitting criteria in a modular Python framework; evaluated nominal 95% interval coverage over 500 simulations per setting and demonstrated criterion-dependent estimates and uncertainty in a cardiac ion-channel application.

Beijing Normal University — Beijing, China
Research Assistant (Advisor: Prof. Dachun Yang) — *Jan. 2022 – Jul. 2024*

Geometry and Characterization of Function Spaces
- Constructed frameworks for representing complex functions in infinite-dimensional spaces; established decomposition, duality, and Fourier characterizations in **two first-author peer-reviewed papers**.

Selected Projects

EasyGrader: AI-Assisted Homework Grading | Live App | Code
- Built and deployed a React/TypeScript and Express grading application with Claude Code, **used during a teaching assistantship**; generated scores and feedback from assignment instructions and custom rubrics.
- Integrated four LLM providers, batch processing, and retries; supported PDF, Jupyter notebook, and code submissions, configurable grading policies, and CSV export for instructor review.

Interactive Option Analytics Application | Live App | Code
- Built and deployed a React/TypeScript application with Vercel serverless functions, integrating Yahoo option-chain data and U.S. Treasury yields for configurable pricing and sensitivity analysis.
- Implemented analytic European-option pricing, a 200-step American-option binomial tree, and Newton–Raphson implied-volatility estimation; provided five option Greeks and an English/Chinese interface.

Selected Publications (2 published, 2 under review)

C. Li and R. Tuo. A Unified Frequentist Framework for Computer Model Calibration. **IISE Transactions** (under review), 2026.
C. Li, X. Zhang, and R. Tuo. Statistical Validation of Computer Models: Global and Subdomain Hypothesis Testing. **Technometrics** (under review), 2026.
C. Li, X. Yan, and D. Yang. Fourier Transform of Anisotropic Hardy Spaces Associated with Ball Quasi-Banach Function Spaces and Its Applications to Hardy–Littlewood Inequalities. **Acta Mathematicae Applicatae Sinica, English Series**, 2024.
C. Li, X. Yan, and D. Yang. Anisotropic Ball Campanato-Type Function Spaces and Their Applications. **Analysis and Mathematical Physics**, 13(3), Article 50, 2023.

Honors & Awards

NeurIPS Financial Assistance Award, full conference and workshop registration — 2026
Selected Oral Presenter, CRSE Workshop, The Wharton School, University of Pennsylvania — 2026
Academic Innovation Award, Beijing Normal University — 2023
National Scholarship, China — 2020
Meritorious Winner, U.S. Mathematical Modeling Competition — 2019
First Prize, Beijing Undergraduate Mathematical Modeling Competition — 2019
Second Prize, Chinese Mathematics Competitions — 2018